

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and trained **Deep Learning networks** for object detection and classification, achieving **92% accuracy** on benchmark datasets by implementing **PyTorch** and **Keras**.
- Selected and processed diverse datasets, including image and sensor fusion data, for optimal **data representation** and model training, integrating with data pipelines to ensure **data integrity**.
- Performed **statistical analysis** and fine-tuning on test results from **deep learning models**, iterating on hyperparameters to improve performance metrics by up to **15%**.
- Collaborated with data engineers and analysts to architect and implement **data acquisition strategies** and robust **data pipelines** for large-scale training sets.
- Applied fundamental **computer vision algorithms** and **machine learning** approaches, including **structure-from-motion** and **localisation & mapping**, to enhance scene understanding.
- Managed software development lifecycles using **version control software** (Git) and operated within a **Linux** environment, contributing to the development of embedded systems.

SKILLS

Deep Learning – PyTorch, Keras, TensorFlow, Neural Network Training, Object Detection, Classification, Structure-from-Motion, Localisation & Mapping

Computer Vision – OpenCV, Image Processing, Sensor Fusion, Online Calibration, Lens Soiling Detection

Programming – Python, Java, C++, Bash Scripting

Data Science & Analytics – Scikit-learn, Pandas, NumPy, Statistical Analysis, Data Modelling, Data Warehousing

Software Development – Linux, Git, Jenkins, Docker, Embedded Systems Development, REST APIs

Business Intelligence – Power BI, Tableau, DAX, Power Query

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin

02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India

01/2021

PROJECTS

Autonomous Vehicle Perception System (HCL Technologies)

- Designed and trained a **deep learning model** using **PyTorch** for real-time **3D object detection** and pedestrian classification, achieving **94% precision** and reducing false positives by **20%**.
- Implemented sensor fusion algorithms to combine data from cameras and LiDAR, enhancing **scene understanding** and trajectory mapping for collision mitigation systems.

- Developed robust **data pipelines** on **Linux** using **Python** and **OpenCV** to process terabytes of automotive sensor data, ensuring high **data integrity** for model training and validation.

Automated Lane Detection and Park-Slot Identification (MSc Data Analytics Project)

- Developed a computer vision pipeline using **Python**, **OpenCV**, and **scikit-learn** to identify lane markings and detect available park slots with **91% accuracy**.
- Applied **machine learning** techniques for image segmentation and classification, optimizing algorithms for performance on simulated automotive embedded platforms.
- Created interactive dashboards in **Power BI** to visualize detection results and performance metrics, enabling stakeholders to review system effectiveness.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)